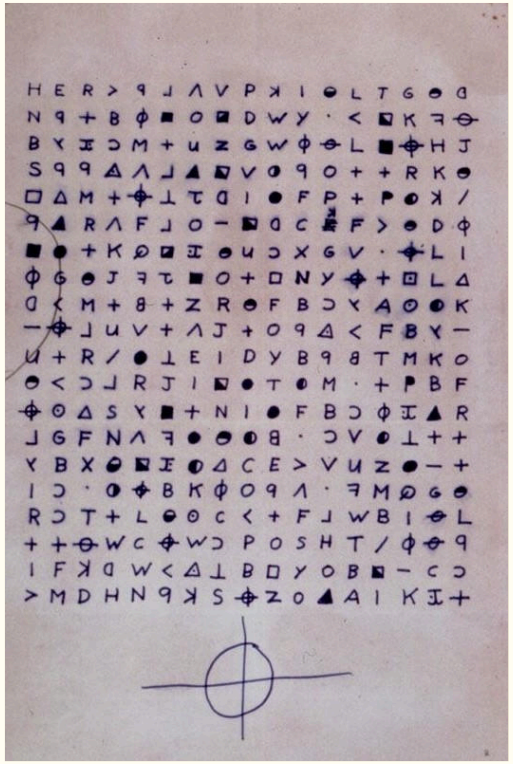
GRADUATE SEMINAR · UCLA

How the Zodiac Z340 Cipher Was Finally Broken



Information theory, Markov chains on groups, and a 51-year-old ciphertext.

Arham Lodha

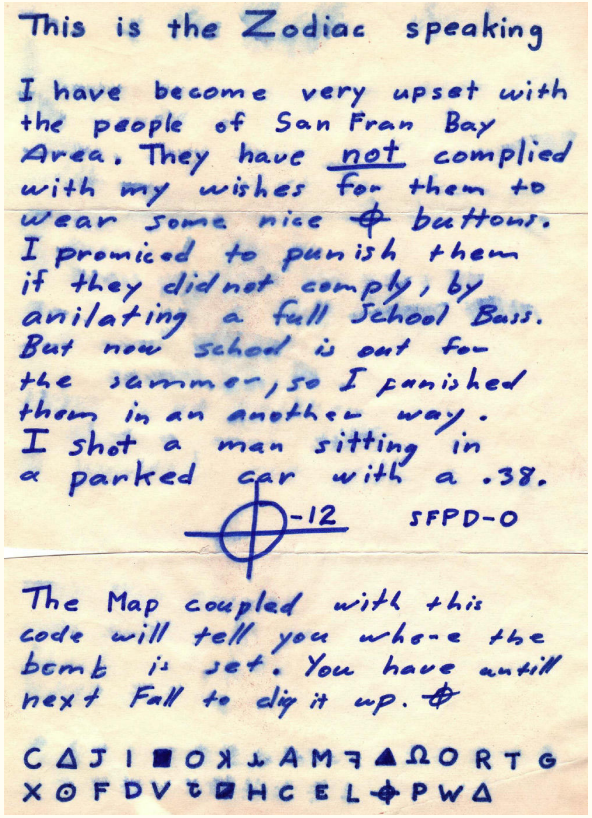
PROLOGUE

The story

San Francisco Bay Area, 1968–1970

The Zodiac

- Dec 1968 – Oct 1969: at least five people murdered around the Bay Area.
- Letters to the *Chronicle*, *Examiner* and *Vallejo Times-Herald* taunting police.
- Four cryptograms, named by length: **Z408**, **Z340**, **Z13**, **Z32**.
- Never identified; the case is still open.


This is the Zodiac speaking

I have become very upset with the people of San Fran Bay Area. They have not complied with my wishes for them to wear some nice ⊕ buttons. I promised to punish them if they did not comply, by anilating a full School Buss. But now school is out for the summer, so I punished them in an another way. I shot a man sitting in a parked car with a .38.

⊕-12 SFPD-0

The Map coupled with this code will tell you where the bomb is set. You have until next Fall to dig it up. ⊕

C △ J I ■ O K ⊥ A M ⅂ ▲ Ω O R T G
X ⊙ F D V ☂ ■ H C E L ⊕ P W △

Z408: solved in a week, by a schoolteacher

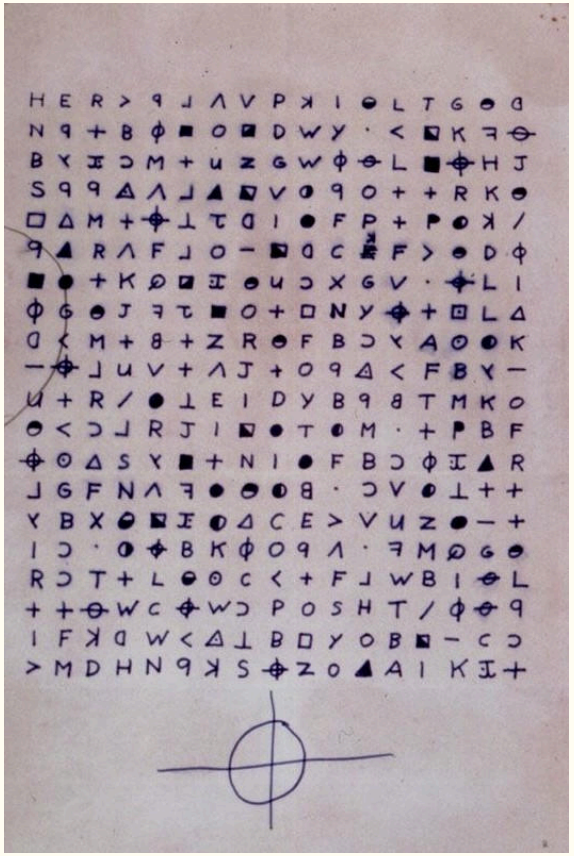


31 July 1969, one of three parts. Print it on the front page, or he kills again.

- Donald and Bettye Harden, Salinas, August 1969.
- Guessed the ego: starts with “I”, contains “KILL”.
- 54 symbols for 26 letters (homophones), but still a substitution.

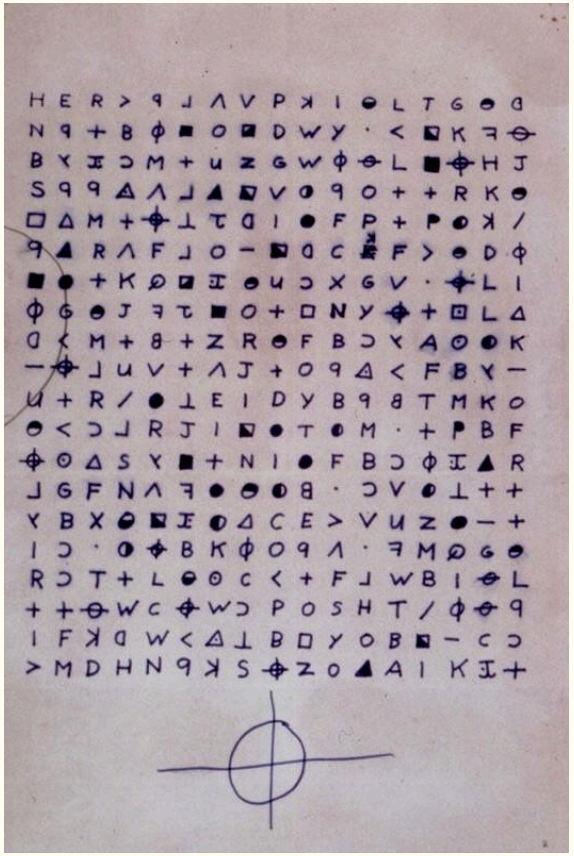
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8 November 1969: Z340



- 340 symbols, 20 rows × 17 columns, **63** distinct symbols.
- Same alphabet of symbols as Z408, so people tried the Z408 key. Nonsense.
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Why? That is the talk.

Z13 and Z32



Z13, April 1970: “My name is —”



Z32, June 1970: “The map coupled with this code will tell you where the bomb is set.”

Plan

1. **Is there a unique answer?** Shannon: entropy, the AEP, unicity distance.
2. **How do we search for the key?** Metropolis on S_{26}, and Diaconis–Shahshahani.
3. **What made Z340 so much harder?** Homophones + transposition.

PART I

Is there a unique answer?

Shannon, 1948–49

How many answers fit?

Z13 repeats only four symbols. Any 13 letters with the same coincidences fit a key:



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STEVE PETE VEST          BARNEY E YEARBY          GENE LYLE LANGE
EDDIE GENE EDEN          ED STEVEN EISEN          E ALVA VANALLEN
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A few of the proposed Z13 solutions: zodiackillerciphers.com/wiki, "Z13 Solutions".

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Every one is **consistent** with the ciphertext. So which is right? Maybe the ciphertext cannot tell. **When does it determine the answer?**

Entropy

X a random variable on a finite set S (letters, cipher symbols). Logs base 2.

$$H(X) = -\sum_{x \in S} \Pr(X = x) \log \Pr(X = x)$$

- X deterministic: $H = 0$. X uniform: $H = \log|S|$, the max.

Markov chains, and language as one

A **Markov chain** on finite S: $\Pr(X_t = j \mid X_0, \ldots, X_{t-1}) = P_{X_{t-1}j}$.

- **Stationary:** $\pi P = \pi$. **Irreducible:** all states communicate. **Aperiodic:** return times have gcd 1.
- **Theorem.** Irreducible + aperiodic $\Rightarrow$ unique stationary π, and $\Pr(X_t = j) \to \pi_j$.
- **Language:** a text is a sample $X_0, X_1, \ldots$ of letters. Bigram model: $P_{ij} = \Pr(j \text{ follows } i)$. The last m letters: still Markov, on S^m.

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Entropy rate: $H = \lim_{n\to\infty} \dfrac{H(X_1, \ldots, X_n)}{n} = -\sum_i \pi_i \sum_j P_{ij} \log P_{ij}$

Cipher systems and redundancy

Shannon: messages $M = S^n$ with the language's distribution; keys $\mathcal{K}$ (uniform); injective $E_k : M \to C$.

- Substitution cipher: $\mathcal{K} = S_{26}$, $\log 26! \approx 88$ bits. Hopeless to brute force… but first ask:

> Given the ciphertext c, is there only one key that produces a **sensible** plaintext?

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- **Redundancy** $D = \log|S| - H$. English: $H \approx 1$ to 1.5, $\log 26 \approx 4.7$, so $D \approx 3$ to 3.7 bits per letter.

The AEP and typical sequences

THEOREM (SHANNON–MCMILLAN–BREIMAN, MARKOV CASE)

X irreducible, aperiodic Markov chain on finite S. Then almost surely

$$-\frac{1}{n}\log \Pr(X_0, X_1, ..., X_n) \longrightarrow H.$$

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Typical set $A_\varepsilon^n = \left\{x : 2^{-n(H+\varepsilon)} < \Pr(x) < 2^{-n(H-\varepsilon)}\right\}$: $\Pr(A_\varepsilon^n) \to 1$ and $|A_\varepsilon^n| \approx 2^{nH}$.

$$\underbrace{2^{nH}}_{\text{typical texts}} \ll \underbrace{2^{n\log|S|}}_{\text{all strings}}$$

English: $2^{1\cdot n}$ vs. $2^{4.7n}$

Unicity distance

A wrong key decrypts to something like a random string: typical with probability $\approx 2^{-nD}$.

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$$n > U = \frac{\log K}{D} \quad \text{(unicity distance)}$$

Past U letters, the key is unique with probability close to 1. Below U, many keys fit and **no amount of cleverness** can pick the right one.

The numbers

Cipher	$\log K$ (bits)	$U \approx \log K / 3$	length
Simple substitution	$\log 26! \approx 88$	≈ 29	a tweet
Z408 (54 symbols)	$\leq 54 \log 26 \approx 254$	≤ 85	408
Z13	$\geq$ huge	$\gg 13$	13
Z32	$\geq$ huge	$\gg 32$	32

- Z408 is $\approx 5 \times$ its unicity distance: the Hardens' answer is **forced**. Slack for misspellings still leaves $U \lesssim 157$.
- Z13, Z32: not unsolved, **unsolvable**.

PART 2

How do we search for the key?

Markov chain Monte Carlo

The Metropolis algorithm

Keys $f : \mathcal{C} \to \mathcal{A}$, far too many to enumerate. Suppose an oracle gives the **plausibility** $\mathrm{Pl}(f)$ of the decoded text.

1. Start with any key f.
2. Pick a random transposition $\sigma \in S_{26}$ and let $f_\sigma = \sigma \circ f$.
3. Move to f_σ with probability $\min(1, \mathrm{Pl}(f_\sigma)/\ \mathrm{Pl}(f))$; otherwise stay at f. Repeat.

$\sigma = \sigma^{-1}$, so $f \to f_\sigma$ is picked as often as $f_\sigma \to f$, and the chain has stationary law $\pi(f) \propto \mathrm{Pl}(f)$.

A plausibility model

Bigram matrix M (26×26), M_{ij} = frequency of j after i in a reference text.

For ciphertext $s_1 s_2 \ldots s_L$:

$$\mathrm{Pl}(f) = \prod_{i=1}^{L-1} M\big(f(s_i), f(s_{i+1})\big)$$

- Smooth M so no entry is 0; work with $\log \mathrm{Pl}$.
- Better model $\Rightarrow$ sharper target $\Rightarrow$ faster search. (Z340 used 5-grams and up.)

An analogous problem: shuffling by transpositions

Drop the plausibility and **always** apply the transposition: the **random transposition walk** on S_n,

$$Q(\sigma) = \begin{cases} \frac{1}{n} & \sigma = e \\ \frac{2}{n^2} & \sigma \text{ a transposition} \\ 0 & \text{otherwise} \end{cases}$$

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For the alphabet, $n = 26$: about $42 + 26c$ transpositions reach every key with nearly equal probability.

Why $\frac{1}{2} n \log n$?

- **Diagonalize:** Q is a class function, so by Schur $\hat{Q}$ acts on each irreducible ρ_λ, $\lambda \vdash n$, as a scalar r_λ. One eigenvalue per partition.
- **Slowest mode:** $\lambda = (n-1, 1)$, with $r = 1 - \frac{2}{n}$ and multiplicity $(n-1)^2$:

$$(n-1)^2 \left(1 - \frac{2}{n}\right)^{2k} \approx n^2 e^{-4k/n} \quad \text{is small iff} \quad k > \frac{1}{2} n \log n.$$

But we don't run that walk

	Random transpositions	Metropolis
Stationary law	U	$\pi(f) \propto \mathrm{Pl}(f)$
Move $f \to \sigma f$	$Q(\sigma)$	$Q(\sigma)\min(1, \pi(\sigma f)/\pi(f))$
Structure	convolution on a group	rejection depends on state

- Not a convolution on the group: Schur no longer diagonalizes the walk, so the proof does not carry over.

Where does π peak?

Ciphertext of length L = English (modeled by M, stationary μ) encrypted by the true key f^*. By the LLN, for any key f:

$$\frac{1}{L}\log \mathrm{Pl}(f) \longrightarrow \sum_{a,b} \mu(a) M(a,b) \log M(f(a), f(b)).$$

- **Gibbs:** largest at $f = f^*$, so $\pi \propto \mathrm{Pl}$ concentrates on f^*, sharper as L grows.

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Tempering $\pi_\beta \propto \mathrm{Pl}^\beta$: $\beta = 0$ is the uniform shuffle (D–S applies), $\beta = 1$ is our algorithm.

A peaked target we can analyze

Mallows model: $\pi_\theta(\sigma) \propto \theta^{d(\sigma,\sigma_0)}$, $0 < \theta \leq 1$, d = Cayley distance. Peaked at σ_0, like Pl at f^*. Same Metropolis chain.

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From σ_0, $k = a(\theta)n\log n + cn$ steps bring the chain within $f(\theta, c) \to 0$ $(c \to \infty)$ of π_θ, where $a(\theta) = \frac{1}{2\theta} + \frac{1}{4\theta}\left(\frac{1}{\theta} - \theta\right)$. Order $n \log n$ steps are also necessary.

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- $\theta = 1$: $a = \frac{1}{2}$, exactly Diaconis–Shahshahani.
- From far away (an n-cycle): order $n^2 \log n$. For $n = 26$: ≈ 85 vs. ≈ 2200 steps.

PART 3

What made Z340 so much harder?

Homophones and a transposition

So it should have been solvable...

Homophonic substitution only:

$$\log K \leq 63 \log 26 \approx 296 \text{ bits}, \quad U \approx \frac{296}{3} \approx 99 \ll 340.$$

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Either the cipher is **more than** homophonic substitution, or the message is too corrupted (typos, misspellings, a hoax) to break.

Errors, or a hidden step?

Clues against corruption:

1. **A spike at spacing 19.** Errors blur patterns; they do not build new ones.
2. **Z408 had a real message:** he could build a working cipher.
3. **A precise 20 × 17 grid:** the layout looks deliberate.
4. **He wanted attention:** a solved cipher means more headlines.

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The extra complication is about the **positions** of the symbols: a **transposition** (suspected by Jeffery Yi, 2014).

Transpositions blow up the key space

Also permute the 340 positions by an **arbitrary** $\sigma \in S_{340}$:

$$\log K \approx 296 + \underbrace{\log(340!)}_{\approx 2374} \approx 2670 \text{ bits}, \quad U \approx \frac{2670}{3} \approx 890 \gg 340.$$

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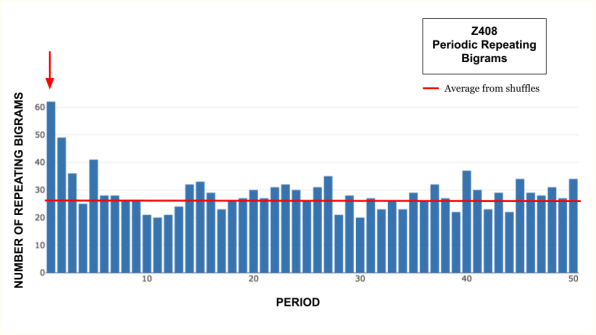
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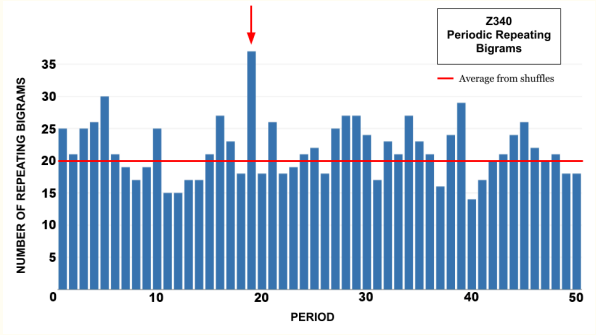
But people pick **simple** permutations: sectioning + a regular transposition + word reversals cost only ≈ 22 extra bits, so $U \approx 318/3 \approx 106$.

The clue: period 19

Repeated pairs (c_i, c_{i+p}) at each spacing p (red: shuffle average).



Z408: peak at $p = 1$ (62).

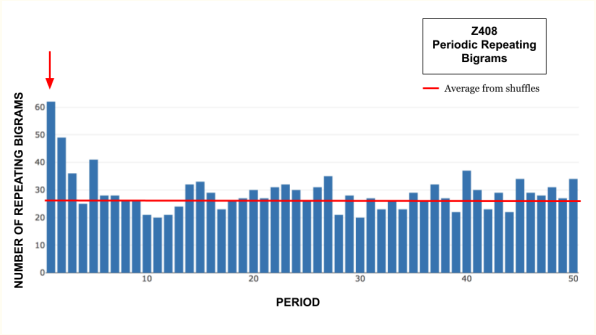


Z340: $p = 1$ is 25; $\boldsymbol{p = 19}$ is 37.

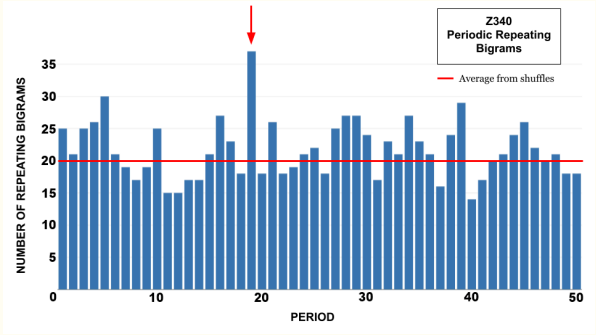
Figure: Oranchak, Blake, Van Eycke, arXiv:2403.17350 (2024).

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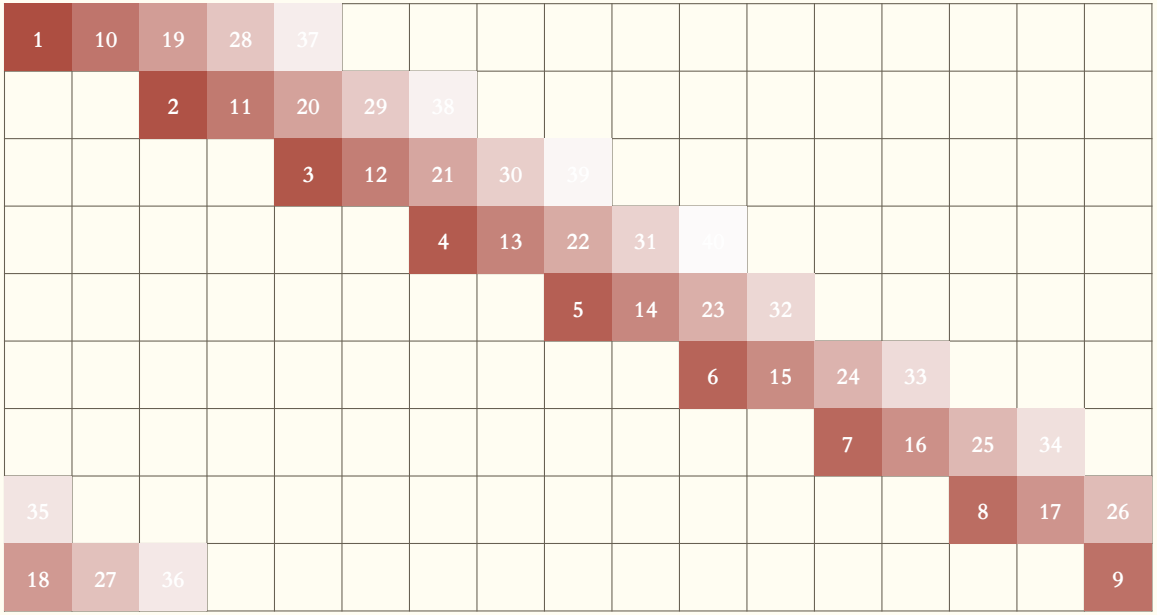
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Figure: Oranchak, Blake, Van Eycke, arXiv:2403.17350 (2024).

At $p = 1$: shuffled Z340 **20**, Z340 as written **25**, Z340-like test ciphers **34.5**. At $p = 19$: **37**.

In reading order Z340 looks almost random; plaintext neighbors were **moved** 19 apart.

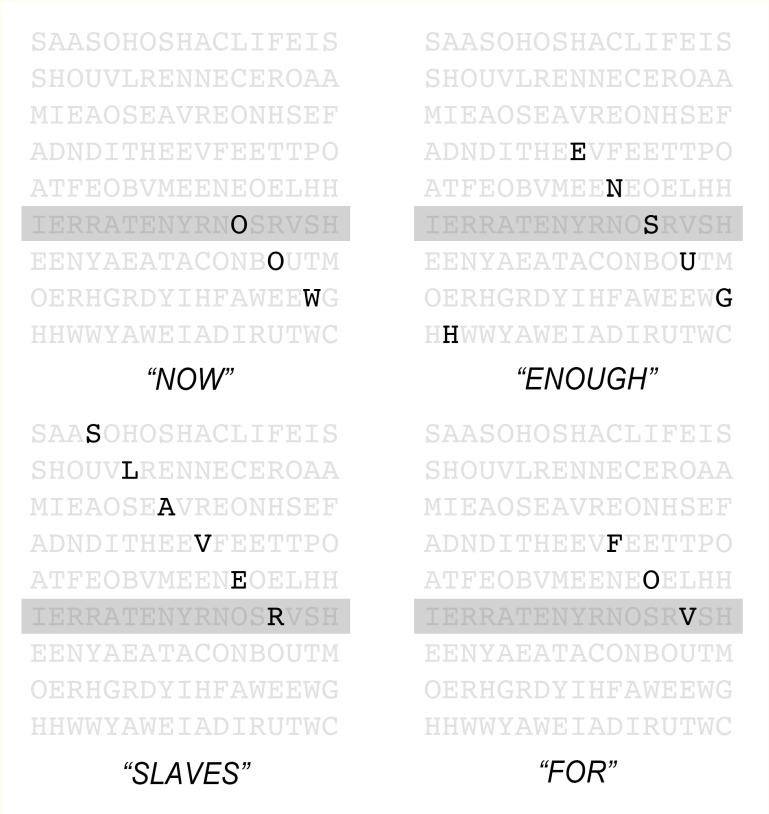
The scheme



Reading order for a 9 × 17 block (first 40 letters).

- Sections: rows 1–9, 10–18, 19–20.
- Sections 1, 2: **one down, two right**, wrapping.
- $17(r+1)+(c+2)$ $= i + 19$.
- "LIFE IS" left out of section 2.
- Rows 19–20: nearly in order.

Encoding slips



- Words garbled on line 15 (row 6 of Section 2): NOW, ENOUGH, SLAVES, FOR.
- Cause: Zodiac skipped a symbol on that line.
- Fix: circularly right-shift RATENYRNOSRVSH within the line.

Errors do happen, but they are local: they garble a line, not the whole message.

Figure: Oranchak, Blake, Van Eycke, arXiv:2403.17350 (2024).

The search, in practice

- **Blake:** generate a large family of plausible transpositions.
- **Van Eycke's AZdecrypt:** for each candidate, solve the homophonic substitution by **simulated annealing** with an **n-gram** model.
- That inner solver is the Metropolis chain from Part 2: higher-order Markov plausibility, tempered over time.
- **Oranchak:** read the outputs and spotted phrases.

The plaintext

```
I HOPE YOU ARE HAVING LOTS OF FUN IN TRYING TO CATCH ME
THAT WASNT ME ON THE TV SHOW
WHICH BRINGS UP A POINT ABOUT ME
I AM NOT AFRAID OF THE GAS CHAMBER
BECAUSE IT WILL SEND ME TO PARADICE ALL THE SOONER
BECAUSE I NOW HAVE ENOUGH SLAVES TO WORK FOR ME …
```

- The "TV show": a caller claiming to be the Zodiac on a talk show, Oct 1969.
- Consistent with Z408's "slaves for my afterlife". No name, of course.

The solution is certified

Key component	bits
Homophonic substitution	296.1
Sectioning (up to 3)	14.7
Transposition + reversals	7.7
Up to 25 letter errors	244.1
Total $\log K$	562.6

J. von zur Gathen, *Unicity distance of the Zodiac-340 cipher*, IACR ePrint 2021/1620.

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$$U \approx 152 < 340$$

$H = 1.36$: $U \approx 132$.
$H = 2.30$: $U \approx 183$.

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$H = 2.30$: $U \approx 183$.

The structure costs 22 bits, not $\log 340! \approx 2374$. Even generous error slack leaves U well under 340: the December 2020 plaintext is **the** solution.

J. von zur Gathen, *Unicity distance of the Zodiac-340 cipher*, IACR ePrint 2021/1620.

Takeaways

1. **Uniqueness first.** Unicity distance $(\log K)/D$ says when a solution can be certified: Z408 yes, Z13 no, Z340 only under a structured transposition.
2. **Search with Markov chains.** Metropolis over keys: order $n \log n$ steps for the uniform (D–S) and a model peaked (D–H) target.
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OPEN QUESTIONS

- No theorem bounds Metropolis mixing for a **realistic** language target.
- Is there a **different** well-specified cipher system, with a similarly small unicity distance, giving another sensible Z340 plaintext?

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